

Investments in Human Capital in the Age of AI

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Abstract

How does Artificial Intelligence reshape human capital investment decisions? I develop a task-based model where risk-averse agents make irreversible training choices based on AI's expected impact and uncertainty. Calibrated to the US labor market, the framework decomposes AI shocks into expected automation, expected augmentation, and their associated uncertainties. The model yields three key results. First, expected automation and its uncertainty drive a structural reallocation, causing workers to systematically avoid training paths facing high automation risk, while choosing paths requiring more manual or social skills. Second, direct AI impact and labor scarcity balance each other; as a result, expected wages rise across all training paths by approximately +14%. Third, automation uncertainty generates an endogenous risk premium of up to +2.3%.

1 Introduction

For most of the twentieth century, technological progress has pointed in the direction of incentivizing investments in human capital. Successive waves of automation primarily displaced routine, mechanical tasks concentrated in low and middle-skill occupations, while simultaneously raising the demand for high-skill workers with higher education (Acemoglu and Autor, 2011; Autor et al., 2003). In this environment, a college degree has been a relatively safe investment, offering high expected earnings and low unemployment risk. The secular rise in the college wage premium and in educational attainment throughout the latter half of the twentieth century is largely a product of this dynamic (Autor, 2019).

Artificial Intelligence (AI) fundamentally disrupts this long-standing relationship. Unlike previous technologies, generative AI demonstrates a high degree of proficiency precisely in the non-routine, cognitive tasks that have historically been the exclusive domain of high-skill workers (Eloundou et al., 2024; Tomlinson et al., 2025; Massenkoff and McCrory, 2026; Webb, 2019). The implications are potentially far-reaching: the

very occupations that attracted generations of students to invest heavily in human capital are now among the most exposed to this new technology.

Importantly, AI’s impact on training choices also has important effects on expected wages. If individuals systematically avoid training paths that are highly exposed to AI, this creates scarcity of workers with those training, which mechanically pushes up their expected wages. Conversely, the inflow of workers into safer training paths creates a congestion of workers that puts downward pressure on wages in those training paths. In this paper, I show that this mechanism has the potential to counteract the direct effects of AI: training paths that are expected to be negatively impacted by AI lose workers, and this scarcity pushes up their expected wages, while training paths that are expected to be positively impacted by AI attract workers, and this congestion pushes down their expected wages. The two forces balance each other, and the final outcome is that expected wages increase for all training paths by approximately +14%.

Another critical feature of this technological breakthrough, however, is its profound uncertainty. The net effect of AI on labor demand is not only unknown, but its expected impact and the magnitude of the uncertainty vary dramatically across occupations and training paths. It is very difficult to predict exactly which tasks will be automated and which will be augmented. Hence, risk-averse individuals may systematically avoid training paths that are relatively uncertain, irrespective of their expected returns. This flight from uncertainty operates through the same scarcity channel: it reduces labor supply in risky paths and generates an endogenous risk premium that pushes their expected wages upward.

To formalize and quantify these dynamics, I develop a task-based model in which heterogeneous, risk-averse agents choose the optimal irreversible human capital investment among J training paths to maximize their expected lifetime utility, net of idiosyncratic investment costs. The key innovation is the decomposition of the AI shock for each training path j into four distinct components: an expected automation effect $\mu_{A,j}$, an expected augmentation effect $\mu_{\gamma,j}$, and their respective variance components $\sigma_{A,j}^2$ and $\sigma_{\gamma,j}^2$, which capture the uncertainty about AI’s ultimate capabilities. The individual discrete choice problem aggregates to a tractable multinomial logit equilibrium.

I calibrate the model to the US labor market. The task-level AI shocks are derived from Althoff and Reichardt (2026), who computed for each task whether it is theoretically automatable with generative AI and an expected augmentation factor if instead generative AI acts as an augmentation technology in that task. Moreover, I rely on the current AI task penetration (Massenkoff and McCrory, 2026), which measures the current AI usage in each task according to Anthropic data. The employment shares across 14 training paths are constructed from the IPUMS American Community Survey (ACS) for 2022, while the median wages are collected from the Bureau of Labor Statis-

tics (BLS). I recover the structural training cost parameters by inverting the pre-AI steady-state equilibrium (2022), and simulate the resulting post-AI equilibrium.

The model delivers three main results, linking theoretical mechanisms to their quantitative implications for the US economy. First, expected automation is the primary driver of labor reallocation, shifting workers away from highly exposed training paths (legal professions, computer science, and business) toward safer alternatives (technical manual jobs and education) or to no training at all. Also, uncertainty about automation amplifies this reallocation: part of the reallocation is driven by pure risk aversion.

Second, the model predicts an across-the-board increase in expected wages by approximately 14%. Training paths very exposed to automation are chosen less (up to -16%); as a result, the scarcity of workers in those training paths mechanically pushes up their expected wages and acts as a countervailing force to the direct negative effect of AI.

Third, when we add uncertainty, it uniquely generates an endogenous risk premium (up to 2.3%). This is a pure general equilibrium effect rather than the standard risk premium arising from investment in a risky asset.

These three mechanisms interact in ways that are difficult to detect using reduced-form methods alone. A highly exposed training path may see its expected wage rise purely due to the labor scarcity, even if AI has a negative direct expected impact. Conversely, a safer path may experience disappointing wage growth due to labor market congestion. This work provides a structural framework to disentangle these intertwined effects.

This paper contributes to three strands of the literature. First, it builds on the large empirical literature on AI’s impact on the labor market (among many others, Acemoglu and Restrepo 2018; Autor and Thompson 2025; Agrawal et al. 2025; Ide and Talamàs 2025; Acemoglu et al. 2025; Gans and Goldfarb 2026), by providing a structural model that allows for a clean decomposition of the supply-side response to the future expected demand shift and uncertainty related to AI. Second, it contributes to the literature on human capital investment under uncertainty (Levhari and Weiss, 1974; Cunha et al., 2005; Cunha and Heckman, 2016), by introducing a new form of risk in the context of human capital investment decisions. Third, it connects to the macro literature on skill-biased technological change (Acemoglu and Autor, 2011; Autor et al., 2003; Autor, 2019) by characterizing the novel mechanism through which AI reshapes economic incentives in ways that are qualitatively different from prior automation waves.

In this regard, the closest paper to this work is Althoff and Reichardt (2026), which provides the foundational task-level framework for how AI simultaneously automates, augments, and simplifies human labor. While they embed these mechanisms into a rich dynamic model of occupational choice, they treat initial skills as exogenous and focus

on skill accumulation on the job. Furthermore, their analysis evaluates the labor market transition under the assumption that uncertainty is symmetric across occupations. This paper takes a complementary approach by explicitly focusing on the endogenous pre-labor market training choice and modeling the training-specific uncertainty surrounding AI's ultimate capabilities.

The rest of the paper is organized as follows. Section 2 develops the theoretical model. Section 3 describes the measurement of the AI shock. Section 4 describes the calibration strategy, and Section 5 presents the quantitative simulation results. Finally, Section 6 concludes.

2 The Model

I consider an economy where agents live for two periods: young ($g = y$) and old ($g = o$). The economy features a finite set of potential training paths. Each training path involves a period of training (when young) followed by a working period (when old). The only exception is the case of no training, which represents a fallback option with no investment in human capital. Training paths may be of two types: a *technical* training path (e.g., plumber, cook) is characterized by a period of apprenticeship as training, while a *cognitive* training path involves a period of studying in college in a certain field before entering the labor market. Let's denote the different training paths by $j \in \{0, 1, \dots, J\}$ ($j = 0$ refers to the case of no training). When young ($g = y$), agents are endowed with a unit of time and choose a training path j . This choice involves an irreversible investment in human capital with uncertain returns. In the working period ($g = o$), agents supply labor inelastically, realize their wage (which depends on the chosen training and labor market conditions), and consume.

2.1 Environment

A representative firm demands labor for each training path. The output per worker for training j is

$$Y_j = \left(\sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau}^{\frac{1}{\rho}} y_{\tau}^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}}$$

where $\mathcal{T}_j$ is the set of relevant tasks for training j , y_{τ} is the output of task τ , and $\theta_{j,\tau}$ is the importance weight of task τ in the production for training j .

The production function for task τ is

$$y_\tau(l_\tau, k_\tau) = \begin{cases} l_\tau \gamma_\tau & \text{if } \tau \in \mathcal{N}_j \\ l_\tau \gamma_\tau + \phi_\tau k_\tau, & \text{if } \tau \in \mathcal{A}_j \end{cases}$$

where $\mathcal{A}_j$ and $\mathcal{N}_j$ are, respectively, the sets of automatable and non-automatable tasks in training j , l_τ is the share of time a worker with training j spends in task τ , and k_τ is the capital per worker in task τ . Finally, γ_τ and ϕ_τ are parameters that capture the productivity of labor and capital in task τ . In this framework, automation refers to an expansion of the set $\mathcal{A}_j$, and labor augmentation refers to an increase in γ_τ .

The problem solved by the representative firm is

$$\begin{aligned} \max_{l_{j,\tau}, k_{j,\tau}} \quad & L_j \left(p_j Y_j - w_j - R \sum_{\tau \in \mathcal{A}_j} k_{j,\tau} \right) \\ \text{subject to } \quad & Y_j = \left(\sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau}^{\frac{1}{\rho}} \tilde{y}_\tau^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}} \\ & \tilde{y}_\tau = y_\tau(l_\tau, k_\tau) \\ & \sum_{\tau \in \mathcal{T}_j} l_{j,\tau} = 1 \end{aligned}$$

where L_j is the share of workers with training j in the labor market. Importantly, L_j is fixed from the previous training period choice, and it is not possible to increase or decrease it until the next period (or next generation).

Assuming that all automatable tasks are produced with capital, i.e., $\frac{w_j}{\gamma_\tau} > \frac{R}{\phi_\tau}$ for all $\tau \in \mathcal{A}_j$, the labor demand for task τ is zero for $\tau \in \mathcal{A}_j$. Thus, the labor demand is given by

$$l_{j,\tau} = \frac{\theta_{j,\tau} \gamma_\tau^{\rho-1}}{\sum_{\tau' \in \mathcal{N}_j} \theta_{j,\tau'} \gamma_{\tau'}^{\rho-1}}$$

and capital per worker is

$$k_{j,\tau} = \theta_{j,\tau} \phi_\tau^{\rho-1} Y_j \left(\frac{p_j}{R} \right)^\rho$$

As a result, the optimal production per worker in training j is

$$Y_j = \Gamma_j^{\frac{\rho}{1-\rho}} \left(\sum_{\tau \in \mathcal{N}_j} \theta_{j,\tau} \gamma_\tau^{\rho-1} \right)^{\frac{1}{\rho-1}}$$

where $\Gamma_j = 1 - \sum_{\tau \in \mathcal{A}_j} \theta_{j,\tau} \left(\frac{R/\phi_\tau}{p_j} \right)^{1-\rho}$ is the labor share for training j .

Finally, since markets are perfectly competitive, the wage is given by

$$w_j = p_j Y_j \Gamma_j = p_j \Gamma_j^{\frac{1}{1-\rho}} \left(\sum_{\tau \in \mathcal{N}_j} \theta_{j,\tau} \gamma_\tau^{\rho-1} \right)^{\frac{1}{\rho-1}} \quad (1)$$

2.2 Equilibrium

Following Althoff and Reichardt (2026), the price of each occupational good p_j is determined in equilibrium through demand and supply. Labor supply is fixed by the previous period's training choice. The workers, in turn, consume and generate demand for the occupational goods. I assume that demand is characterized by a homothetic and invertible demand function $D_j(\{p_j\}_{j=0}^J)$ that maps prices p_j into relative demand for each occupational good.

Definition (Competitive Equilibrium): Given a distribution of workers across training paths $\{L_j\}_{j=0}^J$ such that $\sum_{j=0}^J L_j = 1$, and the supply of capital $\mathcal{K}$, a competitive equilibrium in this economy is given by a sequence of prices $\{p_0, p_1, \dots, p_J, R\}$, such that:

1. Demand for goods equals supply: $D_j(\{p_j\}_{j=0}^J) = L_j Y_j$
2. Demand for capital equals supply: $\mathcal{K} = \sum_{j=0}^J \sum_{\tau \in \mathcal{A}_j} L_j k_{j,\tau}$

2.3 Training Choice

Agents are risk-averse, with preferences described by a CRRA utility function $u(w) = \frac{w^{1-\eta}}{1-\eta}$. In what follows, I will use the log wage, so the equivalent utility function over the log wage is $u(\ln w) = \frac{e^{(1-\eta)\ln w}}{1-\eta}$. Individuals are heterogeneous with respect to a vector of net costs $C_i = [C_{i0}, C_{i1}, \dots, C_{iJ}]$, which aggregates all non-pecuniary disutility and direct expenses specific to each training path. For example, if a worker really hates math, the cost of choosing engineering will be huge, whereas if she really loves literature, the cost of choosing humanities will be very low.

A young agent i chooses the training path j that maximizes the discounted expected utility of future earnings, net of the investment cost:

$$V_i^y(C_i) = \max_j \{ \beta \mathbb{E}_t [u(\ln w_j)] - C_{ij} \} \quad (2)$$

where $\beta \in (0, 1)$ is the discount factor and the log-wage for training path j is given by

Equation 1:

$$\ln w_j = \ln p_j + \frac{1}{1-\rho} \ln \Gamma_j(\mathcal{A}_j) + \frac{1}{\rho-1} \ln \left(\sum_{\tau \in \mathcal{N}_j} \theta_{j,\tau} \gamma_\tau^{\rho-1} \right)$$

where $\Gamma_j = 1 - \sum_{\tau \in \mathcal{A}_j} \theta_{j,\tau} \left(\frac{R/\phi_\tau}{p_j} \right)^{1-\rho}$. As in other standard task-based models, automation has an ambiguous impact because it shrinks the set of tasks performed by humans (displacement effect), but it increases productivity in the remaining tasks due to the cost-savings in the automated tasks (productivity effect); also, it generates a change in the equilibrium prices, which has a direct impact on wages. Similarly, augmentation has an ambiguous impact on wages because it increases productivity of labor, but it changes the equilibrium prices. However, when young, agents do not know what will be the final set of automatable tasks $\mathcal{A}_j$ in each training j , nor which tasks will be augmented. To make this mathematically tractable, I map the set of tasks into a continuum and define a continuous scalar $A_j \in [0, 1]$ representing the share of automated tasks (weighted by importance), and a scalar γ_j representing the average augmentation factor across tasks. Let us suppose their beliefs about the future automation share A_j are well proxied by a normal distribution, $\mathcal{N}(\mu_{A,j}, \sigma_{A,j}^2)$, where $\mu_{A,j}$ is the future expected automation share for training j and $\sigma_{A,j}^2$ captures the uncertainty surrounding future automation in training j . Similarly, for augmentation, workers' beliefs about γ_j are proxied by a normal distribution $\mathcal{N}(\mu_{\gamma,j}, \sigma_{\gamma,j}^2)$, where $\mu_{\gamma,j}$ is the expected augmentation for training j and $\sigma_{\gamma,j}^2$ captures the uncertainty around it. Importantly, all the parameters capturing beliefs are training-specific. Thus, there are some training paths characterized by higher risk of automation and others where augmentation is more likely; also, some training paths could be relatively less subject to both automation and augmentation, but they still can be indirectly impacted through demand spillovers. Moreover, the variance is also training specific and it captures the fact that some training paths are fundamentally more uncertain than others. In the empirical part, I use the existing evidence to calibrate the two moments for each distribution.

Because the log-wage $\ln w_j$ is a non-linear function of both the automated task share A_j and the augmentation factor γ_j , its exact distribution is analytically intractable. To restore tractability, I rely on a first-order Taylor expansion (the Delta method) around the expected shocks. Assuming the automation and augmentation shocks are independent, the linearized log-wage is approximately normally distributed, with expected future log-wage

$$\mathbb{E}[\ln w_j] = \ln p_j(\{L_k, \mu_{A,k}, \mu_{\gamma,k}\}_{k=0}^J) + \frac{1}{1-\rho} \ln \Gamma_j(\mu_{A,j}) + \frac{1}{\rho-1} \ln ((1 - \mu_{A,j}) \mu_{\gamma,j}^{\rho-1})$$

where the endogenous expected price depends on the reallocation of labor supply, and expected automation and augmentation; the expected labor share in each training j depends on expected automation; finally, the last term captures the expected productivity, which depends on expected automation and on the share of tasks that will not be automated. The overall variance is the sum of the variances of the underlying shocks, weighted by the squared sensitivities:

$$Var(\ln w_j) = \left(\frac{\partial \ln w_j}{\partial A_j} \bigg|_{\mu_{A,j}, \mu_{\gamma,j}} \right)^2 \sigma_{A,j}^2 + \left(\frac{\partial \ln w_j}{\partial \gamma_j} \bigg|_{\mu_{A,j}, \mu_{\gamma,j}} \right)^2 \sigma_{\gamma,j}^2$$

Given this normal approximation for the log-wage, the expected utility can be evaluated exactly using the moment-generating function of the normal distribution:

$$\mathbb{E}[u(\ln w_j)] = \frac{1}{1-\eta} \exp \left((1-\eta)\mathbb{E}(\ln w_j) + \frac{(1-\eta)^2}{2} Var(\ln w_j) \right)$$

The certainty equivalent Ω_j is defined as the guaranteed wage that yields the same expected utility, $u(\Omega_j) = \mathbb{E}[u(\ln w_j)]$. Equating the two yields:

$$\frac{e^{(1-\eta)\Omega_j}}{1-\eta} = \frac{1}{1-\eta} \exp \left((1-\eta)\mathbb{E}(\ln w_j) + \frac{(1-\eta)^2}{2} Var(\ln w_j) \right)$$

Canceling the constant term, taking logs, and rearranging provides the exact certainty equivalent for training path j :

$$\Omega_j = \underbrace{\ln p_j + \frac{1}{1-\rho} \ln \Gamma_j + \frac{1}{\rho-1} \ln ((1-\mu_{A,j})\mu_{\gamma,j}^{\rho-1})}_{\mathbb{E}(\ln w_j)} - \underbrace{\frac{\eta-1}{2} Var(\ln w_j)}_{\text{Risk premium}}$$

which formalizes exactly how agents penalize uncertainty depending on their degree of risk aversion η . Importantly, this model creates a training-specific risk premium that is linked with the relative uncertainty regarding future automation and augmentation due to AI in each training path.

Substituting $\mathbb{E}[u(\ln w_j)]$ with $u(\Omega_j)$, the agent's objective function becomes:

$$V_i^y(C_i) = \max_j \{ \beta u(\Omega_j) - C_{ij} \}$$

To render the discrete choice problem analytically tractable, I apply a first-order Taylor expansion to the utility of the certainty equivalent $u(\Omega_j)$ around a baseline average log-wage $\bar{\mu}$. Substituting $u(\Omega_j) \approx u(\bar{\mu}) + u'(\bar{\mu})(\Omega_j - \bar{\mu})$ into the objective, the

maximization problem simplifies to:

$$\max_j \{ \beta \Omega_j - C_{ij}^* \} \quad (3)$$

where $C_{ij}^* \equiv C_{ij}/u'(\bar{\mu})$ is the normalized cost parameter. Notice that, although we are linearizing the utility function, the risk aversion coefficient η still enters into the decision rule through the risk premium in the certainty equivalent Ω_j .

I assume that the normalized cost is determined by a training-specific common average cost, $\bar{C}_j^*$, and an idiosyncratic preference component, such that $C_{ij}^* = \bar{C}_j^* - \epsilon_{ij}$. Assuming the idiosyncratic shocks ϵ_{ij} are independently and identically distributed following a Type I Extreme Value (Gumbel) distribution with location parameter zero and scale parameter ζ , the aggregate market share of labor choosing training path j is mapped by the standard multinomial conditional logit formulation (McFadden (1972)):

$$L_j^* = \frac{\exp((\beta \Omega_j - \bar{C}_j^*)/\zeta)}{\sum_{h=0}^J \exp((\beta \Omega_h - \bar{C}_h^*)/\zeta)} \quad (4)$$

2.4 Theoretical Results and Equilibrium Properties

Using the structural discrete choice framework derived above, this section establishes the comparative statics of the model with respect to the four parameters of the agents' beliefs about the automation and augmentation shocks: the expected levels ($\mu_{A,j}$ and $\mu_{\gamma,j}$) and their respective uncertainties ($\sigma_{A,j}^2$ and $\sigma_{\gamma,j}^2$).

Proposition 1 (Downward-Sloping Labor Demand). *Because output per worker Y_j is independent of L_j , an increase in the labor supply L_j for training path j raises the aggregate supply of occupational good j . Given the homothetic demand function D_j , this generates a fall in the equilibrium price p_j , which translates one-for-one into a fall in the expected log-wage:*

$$\frac{\partial \mathbb{E}[\ln w_j]}{\partial L_j} = \frac{\partial \ln p_j}{\partial L_j} < 0 \quad (5)$$

Conversely, a decrease in L_j raises the equilibrium price p_j and therefore raises $\mathbb{E}[\ln w_j]$. Expected wages are thus strictly decreasing in labor supply: the labor demand curve is downward sloping.

Intuition: When more workers crowd into a training path, they collectively produce more of the associated occupational good. The resulting excess supply depresses its price, and since wages equal the value of marginal product, expected wages fall. The reverse holds when workers leave a path: scarcity drives up the price and lifts expected wages. This mechanism is the central general-equilibrium channel through which all

belief shocks (expected automation, expected augmentation, and their uncertainties) translate into wage adjustments.

This allows us to map how exogenous changes in beliefs affect actual equilibrium wages. Note that the derivatives derived from the conditional logit formula in Proposition 2 represent the *direct* (or partial equilibrium) shifts in labor supply holding expected wages and prices constant. In general equilibrium, these initial shifts trigger price adjustments that ultimately dampen the final reallocation; however, this effect can never reverse the direction of the response.

Proposition 2 (Flight from Uncertainty). *An increase in automation uncertainty ($\uparrow \sigma_{A,j}^2$) or augmentation uncertainty ($\uparrow \sigma_{\gamma,j}^2$) for training path j leads to a decrease in the equilibrium employment L_j^* :*

$$\frac{\partial L_j^*}{\partial \sigma_{A,j}^2} = -\frac{\beta(\eta-1)}{2\zeta} \left(\frac{\partial \ln w_j}{\partial A_j} \right)^2 L_j^*(1-L_j^*) < 0 \quad (6)$$

$$\frac{\partial L_j^*}{\partial \sigma_{\gamma,j}^2} = -\frac{\beta(\eta-1)}{2\zeta} \left(\frac{\partial \ln w_j}{\partial \gamma_j} \right)^2 L_j^*(1-L_j^*) < 0 \quad (7)$$

Intuition: As a career path becomes more unpredictable due to exposure to AI, risk-averse individuals systematically flee to safer alternatives. The magnitude of this structural flight depends directly on their degree of risk aversion (η).

Proposition 3 (Endogenous Risk Premium). *An increase in either automation or augmentation uncertainty for training path j ($\uparrow \sigma_{A,j}^2, \uparrow \sigma_{\gamma,j}^2$) raises the expected wages $\mathbb{E}(\ln w_j)$. This reflects an endogenous risk premium driven by labor scarcity:*

$$\frac{\partial \mathbb{E}(\ln w_j)}{\partial \sigma_{A,j}^2} = \underbrace{\frac{\partial \mathbb{E}(\ln w_j)}{\partial p_j}}_{(>0)} \underbrace{\frac{\partial p_j}{\partial L_j}}_{(<0)} \underbrace{\frac{\partial L_j^*}{\partial \sigma_{A,j}^2}}_{(<0)} > 0, \quad \frac{\partial \mathbb{E}(\ln w_j)}{\partial \sigma_{\gamma,j}^2} = \underbrace{\frac{\partial \mathbb{E}(\ln w_j)}{\partial p_j}}_{(>0)} \underbrace{\frac{\partial p_j}{\partial L_j}}_{(<0)} \underbrace{\frac{\partial L_j^*}{\partial \sigma_{\gamma,j}^2}}_{(<0)} > 0 \quad (8)$$

Intuition: This characterizes one of the core theoretical mechanisms of the model. Uncertainty reduces the share of labor choosing training path j (see Proposition 2). The resulting labor scarcity pushes up the equilibrium price of the occupational good, mechanically increasing the realized wage for those who remain.

3 Measuring the AI Shocks

To implement the model quantitatively, we need to measure the expected values ($\mu_{A,j}$ and $\mu_{\gamma,j}$) and the uncertainties ($\sigma_{A,j}^2$ and $\sigma_{\gamma,j}^2$) of the automation and augmentation shocks for each training path j . I compute these moments by categorizing the tasks comprising each training path based on their current and potential AI exposure.

This categorization relies on two main sources: the actual task penetration of AI (Massenkoff and McCrory, 2026), and the theoretical automatability and augmentation potential of tasks (Althoff and Reichardt, 2026). The task penetration is zero if, at the moment, AI is not used in that task. If instead AI is already used in that task, its penetration is greater than or equal to $1/2$, with values closer to 1 meaning that AI is more automating than augmenting that task (see Massenkoff and McCrory (2026) for more details). Theoretical automatability is either 1 if the task is potentially automatable or 0 otherwise. Theoretical augmentation is 1 if the task is not augmentable and greater than 1 if it is potentially augmentable, for example a value of 1.1 means that AI can boost labor productivity by 10% in that task (see Althoff and Reichardt (2026) for more details).

First, I define the *current status* of a task:

- **Automated:** The actual task penetration is greater than or equal to 0.9 and the task is automatable (automatable = 1).
- **Augmented:** The actual task penetration is greater than 0, the task is not automatable (automatable = 0). Alternatively, the task is automatable (automatable = 1), but the current task penetration is less than 0.9.
- **Neither:** If the actual task penetration is zero, the task is currently neither automated nor augmented.

Second, I define the *potential status* of a task:

- If a task is already automated, its potential status is automated (with no uncertainty).
- If a task is already augmented and it is not automatable, its potential status is augmented (with no uncertainty).
- **Potentially automatable:** The task’s actual penetration is lower than 0.9, but the task is theoretically automatable (automatable = 1).
- **Potentially augmentable:** The task’s actual penetration is zero, it is not automatable (automatable = 0), and its augmentation factor is greater than 1.
- **Not affected by AI:** The task’s actual penetration is zero, it is not automatable (automatable = 0), and its augmentation factor is exactly 1.

To formalize this aggregation, let $\mathcal{T}_j$ be the set of tasks associated with training path j , and let $\theta_{j,\tau}$ denote the importance weight of task τ (such that $\sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau} = 1$). I treat the ultimate AI exposure of each task as a random variable. Let $a_\tau \in \{0, 1\}$ be

an indicator of whether task τ will eventually be automated, and let $b_\tau \in \{0, 1\}$ be an indicator of whether it will be augmented.

For automation, if a task is already automated, its future outcome is certain: $\mathbb{E}[a_\tau] = 1$ and $\text{Var}(a_\tau) = 0$. Conversely, if it is not even potentially automatable, $\mathbb{E}[a_\tau] = 0$ and $\text{Var}(a_\tau) = 0$. The uncertainty lies entirely with the tasks that are *potentially automatable* but not yet automated. For these tasks in the “gap”, I assume a probability $p_A \in (0, 1)$ that the theoretical automatability translates into actual automation. Thus, for tasks in this gap, $\mathbb{E}[a_\tau] = p_A$ and $\text{Var}(a_\tau) = p_A(1 - p_A)$.

The expected automation share $\mu_{A,j}$ for training path j is the weighted sum of these task-level expectations:

$$\mu_{A,j} = \sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau} \mathbb{E}[a_\tau] = S_{A,j}^{cur} + p_A (S_{A,j}^{pot} - S_{A,j}^{cur}) \quad (9)$$

where $S_{A,j}^{cur}$ is the currently automated (weighted) share and $S_{A,j}^{pot}$ is the maximum potentially automatable (weighted) share. To formalize the possibility that AI success is correlated across tasks within the same training path, I introduce a correlation parameter $\kappa \in [0, 1]$. A value of $\kappa = 0$ implies independent task automation (maximizing diversification), whereas $\kappa = 1$ implies perfect correlation (AI automates either all potential tasks or none). The variance of the automation shock then decomposes into a convex combination of the independent and perfectly correlated variances:

$$\sigma_{A,j}^2 = p_A(1 - p_A) \left[(1 - \kappa) \sum_{\tau \in \text{Gap}_{A,j}} \theta_{j,\tau}^2 + \kappa \left(\sum_{\tau \in \text{Gap}_{A,j}} \theta_{j,\tau} \right)^2 \right] \quad (10)$$

where $\text{Gap}_{A,j}$ is the set of tasks that are potentially automatable but not yet automated. First, notice that the total variance is strictly increasing in the total automation gap $\sum_{\tau \in \text{Gap}_{A,j}} \theta_{j,\tau}$; therefore, training paths with a higher unrealized potential for automation inherently face higher uncertainty. Second, the term $\sum \theta_{j,\tau}^2$ acts as a Herfindahl-Hirschman Index (HHI) of task concentration. This implies that, especially when $\kappa < 1$, holding the gap share constant, training paths that are highly specialized in a few uncertain tasks face significantly more risk than paths with a diversified portfolio of many small tasks.

A symmetric logic applies to augmentation. Let γ_τ be the theoretical augmentation factor for task τ (where $\gamma_\tau > 1$ means the task is augmentable, and $\gamma_\tau = 1$ means no gain). For tasks that are strictly augmented (already augmented and *not* potentially automatable), their outcome is certain: $\mathbb{E}[b_\tau] = 1$ and $\text{Var}(b_\tau) = 0$. Conversely, if the task is not even potentially augmentable, $\mathbb{E}[b_\tau] = 0$ and $\text{Var}(b_\tau) = 0$.

The uncertainty in augmentation, captured by the set $\text{Gap}_{\gamma,j}$, arises from two dis-

tinct sources. First, tasks that are *potentially augmentable* but not yet augmented have a probability p_γ of realizing this productivity boost. Second, tasks that are *already augmented* but also *potentially automatable* face the risk of full automation; if automation succeeds (with probability p_A), the human is replaced and the augmentation factor effectively drops back to 1. Thus, they only have a probability $1 - p_A$ of remaining augmented. Assuming for simplicity that $p_\gamma = 1 - p_A \equiv p$ (as implemented in the baseline calibration where both probabilities are 0.5), both sources of uncertainty yield an expected indicator $\mathbb{E}[b_\tau] = p$ and a variance $\text{Var}(b_\tau) = p(1 - p)$. The expected augmentation factor and its variance for training path j are then:

$$\mu_{\gamma,j} = \sum_{\tau \in T_j} \theta_{j,\tau} (1 + \mathbb{E}[b_\tau](g_\tau - 1)) \quad (11)$$

$$\sigma_{\gamma,j}^2 = p_\gamma(1 - p_\gamma) \left[(1 - \kappa) \sum_{\tau \in \text{Gap}_{\gamma,j}} \theta_{j,\tau}^2 (g_\tau - 1)^2 + \kappa \left(\sum_{\tau \in \text{Gap}_{\gamma,j}} \theta_{j,\tau} (g_\tau - 1) \right)^2 \right] \quad (12)$$

Just as with automation, this variance is driven by two key factors: the size of the potential augmentation gap (meaning paths with a higher potential productivity boost naturally face higher uncertainty) and the concentration of that potential across a few specific tasks.

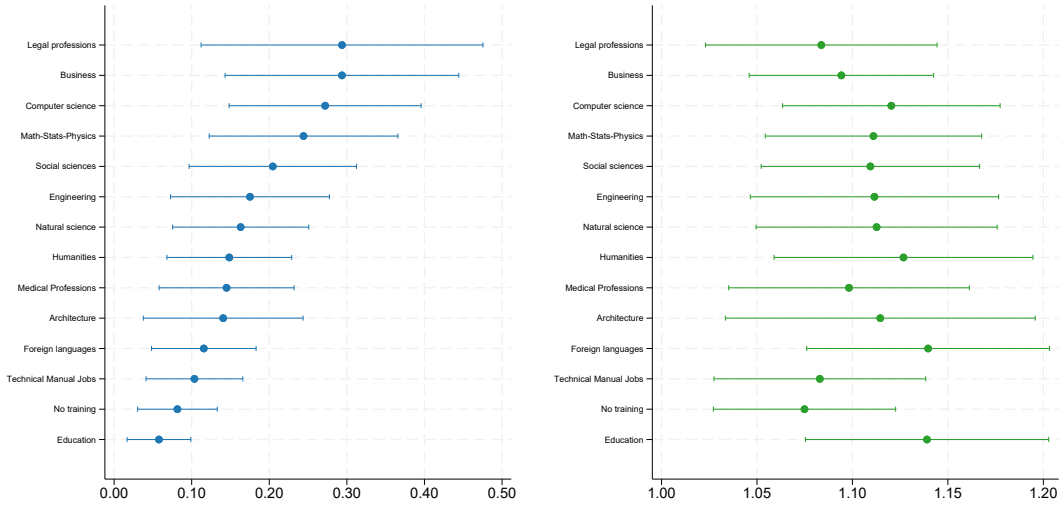


Figure 1: Expected Automation and Augmentation across Training Paths (with 1 standard deviation confidence interval).

Figure 1 displays the expected automation ($\mu_{A,j}$) and expected augmentation ($\mu_{\gamma,j}$) shocks across different training paths. Furthermore, the confidence intervals capture differences in uncertainty across training paths. For instance, the training paths with the most automation uncertainty are legal professions and business, because many of

their tasks are potentially automatable, but not yet automated. Computer science has a lower uncertainty compared to them (still relatively high) because some of its tasks have already been automated. On the other hand, paths like technical manual jobs and education have a very low expected automation (and uncertainty) because their tasks are more difficult to automate. As far as augmentation is concerned, both expectations and uncertainty are similar across training paths; as a result, its role in shaping labor reallocation is limited compared to automation.

4 Calibration

The quantitative implementation of the model proceeds in two main stages. First, I calibrate the model to a pre-AI baseline equilibrium (representing the year 2022) where AI shocks are not yet anticipated. This allows me to recover the fundamental structural parameters of the economy, particularly the unobserved training costs. Second, I introduce the measured AI shocks (both expected productivity and uncertainty) derived in Section 3 to simulate the new post-AI equilibrium and decompose its effects on employment and wages.

4.1 Pre-AI Baseline Equilibrium (2022)

In the pre-AI steady state, agents do not anticipate the arrival of Artificial Intelligence. Consequently, their expectations about future automation and augmentation are zero ($\mu_{A,j} = \mu_{\gamma,j} = 0$), and there is no uncertainty regarding these shocks ($\sigma_{A,j}^2 = \sigma_{\gamma,j}^2 = 0$). Without uncertainty, the risk premium is zero, and the certainty equivalent Ω_j collapses exactly to the expected current log-wage: $\Omega_j^{pre} = \ln w_j^{2022}$.

Using the optimal labor supply derived in Equation (4), the employment share for training path j in the pre-AI equilibrium is given by:

$$L_j^{2022} = \frac{\exp((\beta \ln w_j^{2022} - \bar{C}_j^*)/\zeta)}{\sum_{h=0}^J \exp((\beta \ln w_h^{2022} - \bar{C}_h^*)/\zeta)} \quad (13)$$

where L_j^{2022} and w_j^{2022} are the observed employment shares and median wages from the 2022 American Community Survey (ACS) and Bureau of Labor Statistics (BLS) data, across the 14 training paths.

By observing the actual labor allocations and wages, I can invert this mapping to exactly recover the unobserved normalized training costs $\bar{C}_j^*$. Normalizing the cost of the no-training option ($j = 0$) to zero ($\bar{C}_0^* = 0$), the structural cost parameters are identified as:

$$\bar{C}_j^* = \beta(\ln w_j^{2022} - \ln w_0^{2022}) - \zeta(\ln L_j^{2022} - \ln L_0^{2022}) \quad (14)$$

On the labor demand side, following the procedure outlined in Althoff and Reichardt (2026), I assume that occupational goods are substituted with a constant elasticity of substitution (CES) ϵ , so that demand for occupation j satisfies $D_j(\mathbf{p}) \propto \alpha_j p_j^{-\epsilon}$. I set the elasticity of substitution between occupations $\epsilon = 1.57$ —the midpoint between 1.81 (Burstein et al., 2019) and 1.34 (Caunedo et al., 2023). The CES importance weights α_j are then identified from observed occupational wage shares and the estimated occupational prices because for any two occupations i and j :

$$\frac{\alpha_i}{\alpha_j} = \left(\frac{p_i}{p_j}\right)^{\epsilon-1} \times \frac{\text{Wage share of occupation } i}{\text{Wage share of occupation } j} \quad (15)$$

Because no tasks are automated in the pre-AI baseline economy, the labor share is exactly one ($\Gamma_j = 1$) and output per worker is normalized to one ($Y_j = 1$). Consequently, the competitive wage equation $w_j = p_j Y_j \Gamma_j$ simplifies to $p_j = w_j$. Thus, I can directly substitute the observed 2022 wages w_j^{2022} for the prices p_j in the equation above to perfectly calibrate the demand shifters α_j .

4.2 Externally Calibrated Parameters

To simulate the model quantitatively, I calibrate several fundamental parameters directly from the recent literature on task-based models, relying primarily on the estimates provided by Althoff and Reichardt (2026).

Macro Elasticities and Preferences: I set the elasticity of substitution across tasks within an occupation to $\rho = 0.49$, implying that tasks are complementary. For the agents’ preferences, the annual discount factor is set to $\beta = 0.813$, corresponding to an annual discount rate of 3% and considering 7 years (from 18 to 25) for the training period, and the coefficient of relative risk aversion is set to a baseline value of $\eta = 5$. In the results section, I additionally test the sensitivity of the model to a conservative scenario ($\eta = 2$) and a high risk aversion scenario ($\eta = 10$). Finally, following Althoff and Reichardt (2026), the Gumbel dispersion parameter for idiosyncratic occupational preferences is set to $\zeta = 0.053$.

Task Exposure and Augmentation: The expected shocks (μ) and uncertainties (σ^2) defined in Section 3 require calibrating the probability that a task in the “gap” is eventually exposed to AI. To reflect a state of maximum agnostic uncertainty regarding the future of these tasks, I set the adoption probabilities to $p_A = 0.5$ and $p_\gamma = 0.5$. Furthermore, I use the task-specific augmentation factors g_τ from (Althoff and Reichardt, 2026).

Capital-Labor Substitution: While the pre-AI baseline economy assumes no tasks are automated by generative AI (so the initial labor share is exactly $\Gamma_j^{pre} = 1$),

simulating the post-AI economy requires quantifying how the labor share responds to the automation shock. Rather than calibrating task-specific capital rental rates and productivities, I follow Althoff and Reichardt (2026) and assume that the cost savings of using AI instead of labor for automatable tasks is 27%, yielding a relative cost of capital $\chi = 0.73$. Using their structural derivation, the post-AI labor share for any training path j can be approximated as:

$$\Gamma_j^{post} \approx \frac{\chi^{\rho-1} (1 - \mu_{A,j})}{\mu_{A,j} + \chi^{\rho-1} (1 - \mu_{A,j})} \quad (16)$$

where $\mu_{A,j}$ is the expected total share of tasks automated by AI in training path j (as derived in Section 3).

4.3 Simulating the Post-AI Equilibrium

Once the structural parameters are pinned down, I introduce the AI shocks. For each training path j , I plug in the expected values $(\mu_{A,j}, \mu_{\gamma,j})$ and variances $(\sigma_{A,j}^2, \sigma_{\gamma,j}^2)$ computed in Section 3.

These shocks alter the certainty equivalent Ω_j^{post} for each training path through both the direct expected productivity shifts and the endogenous risk premiums. The new post-AI equilibrium is defined as the fixed point where:

1. Risk-averse agents optimally reallocate across training paths based on the new certainty equivalents, forming the post-AI labor supply L_j^{post} .
2. The new labor supply alters the production of occupational goods, which in turn adjusts the equilibrium prices p_j^{post} according to the CES demand function.
3. Expected log-wages $\mathbb{E}[\ln w_j]^{post}$ adjust endogenously to clear the market.

By comparing the post-AI equilibrium to the pre-AI baseline, we can quantify the total effect of AI on employment and expected wages, and further decompose these effects into the demand channel, the uncertainty channel, and the crowding effect.

5 Results and Decomposition

This section presents the quantitative predictions of the calibrated model. To isolate the contribution of each economic mechanism, the full post-AI equilibrium is decomposed sequentially into four channels: the *expected automation* effect, the *expected augmentation* effect, and the *uncertainty* effects associated with both automation and augmentation shocks.

5.1 Employment Reallocation and the Flight to Safety

Figure 2 presents the decomposition of AI-induced employment shifts across the 14 training paths.

The *expected automation* channel drives the primary structural transformation. As AI substitutes for cognitive tasks, labor demand shifts away from highly automatable cognitive paths such as business, computer science, and legal professions. Conversely, paths with high physical or interpersonal requirements—such as technical manual jobs and education, as well as the “no training” residual category, experience positive labor inflows. The *expected augmentation* channel plays a comparatively minor role in employment reallocation, as human-enhancing technologies boost output across most sectors relatively symmetrically without dramatically shifting relative labor demand.

Crucially, the *uncertainty* over automation triggers a secondary reallocation. Risk-averse agents actively flee the highly uncertain training paths (e.g., business and legal professions). This flight from uncertainty amplifies the initial expected automation shock, accelerating the employment drop in exposed careers and crowding workers into the perceived safer options. Uncertainty regarding augmentation has virtually zero impact on employment, again because the augmentation uncertainty is relatively similar across training paths.

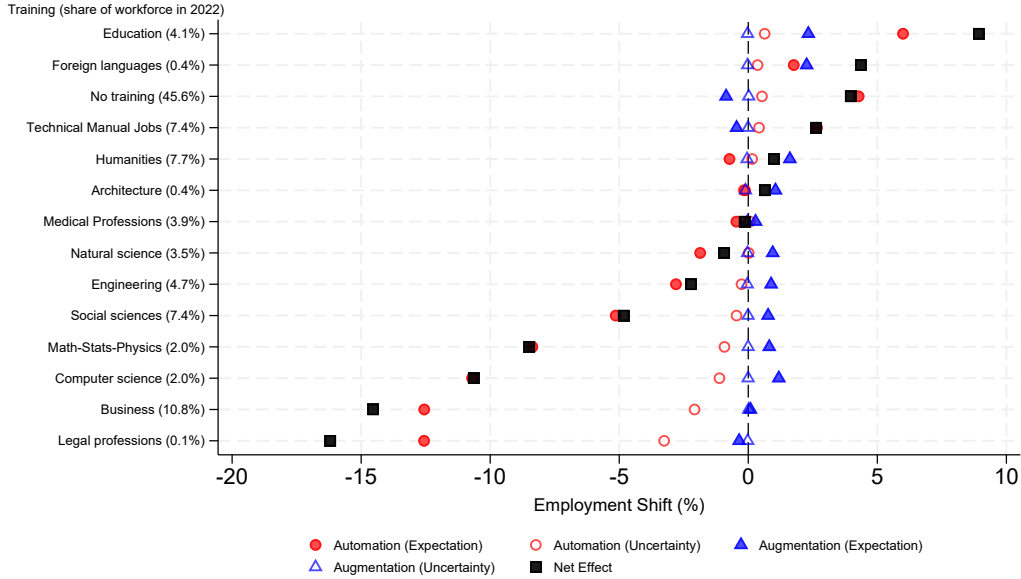


Figure 2: Decomposition of AI-induced employment shifts across training paths. The total net effect (black squares) is decomposed into four channels: expected automation (red circles), uncertainty over automation (red hollow circles), expected augmentation (blue triangles), and uncertainty over augmentation (blue hollow triangles). Shifts are expressed as percentage changes relative to the pre-AI baseline share.

5.2 Wage Effects and the Endogenous Risk Premium

The wage decomposition (Figure 3) reveals a starkly different pattern, underscoring the power of general equilibrium forces. The expected AI shocks—both automation and augmentation—are unequivocally beneficial for aggregate productivity, generating an across-the-board increase in expected wages ranging from +4.2% to +5.4% from automation, and +9.5% from augmentation. The primary driver of this result is the reallocation of labor across training paths. If we take labor supply as fixed to the pre-AI baseline, the expected wage impact is dramatically diverse. Training paths very exposed to automation (legal, business, and computer science) would see their expected wages drop by more than 2%, whereas the safer paths (technical manual jobs, education, and no training) would gain up to +11%. The fact that individuals reallocate toward positively impacted paths, however, erodes the expected wage gains in those paths and generates a scarcity effect in the exposed paths. These results highlight the importance of accounting for the labor supply responses when evaluating the impact of AI on wages.

The impact of uncertainty is qualitatively similar. Risk-averse agents flee the highly uncertain training paths, which reduces employment in those paths. The labor scarcity generated by this flight to safety, however, creates an endogenous risk premium that drives up expected wages in the exposed paths. Quantitatively, this channel is less pronounced, but it is still economically significant. The most uncertain training paths (legal and business) see their expected wages increase by +2.1% and +1.3%, respectively, only because of the uncertainty-driven labor scarcity.

To further highlight this endogenous risk premium, Figure 4 plots the expected wage changes against the variance of the AI shock across the different training paths. The evident positive relationship confirms that paths facing higher technological uncertainty compensate workers with higher expected wages. This highlights the general equilibrium effect: as risk-averse workers flee highly volatile training paths, the resulting labor scarcity mechanically drives up the expected wages in those training paths.

6 Conclusion

This paper studies how Artificial Intelligence reshapes human capital investment when its impact is uncertain and heterogeneous across training paths. I develop a task-based model in which risk-averse agents make irreversible training choices to maximize a certainty equivalent that penalizes AI-related wage variance, and I calibrate the framework to the US labor market using ACS and BLS data together with task-level automation and augmentation exposure from Althoff and Reichardt (2026) and Massenkoff and McCrory (2026).

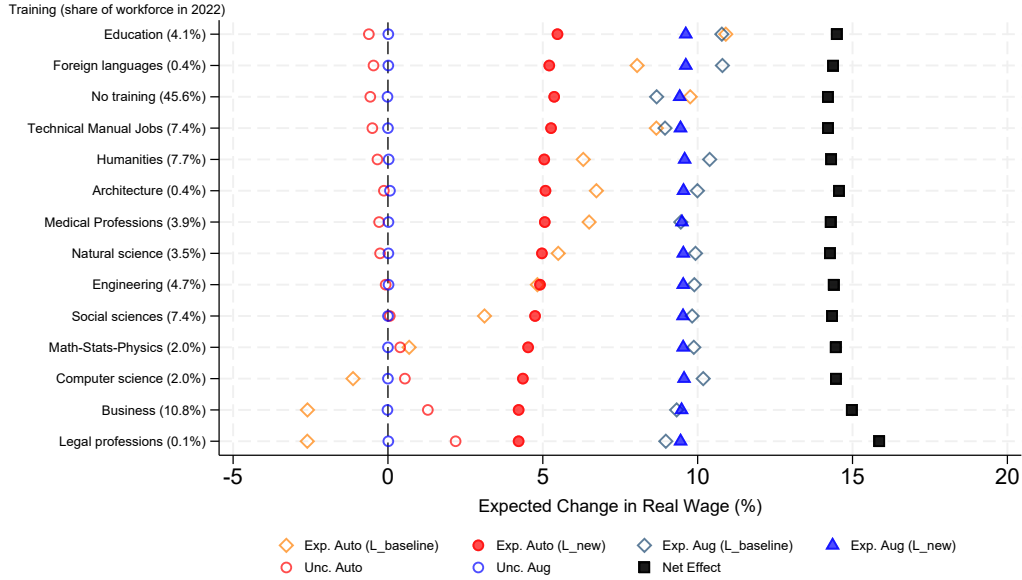


Figure 3: Decomposition of AI-induced expected wage changes across training paths. The total net wage change (black squares) is decomposed into four channels: expected automation (red circles), uncertainty over automation (red hollow circles), expected augmentation (blue triangles), and uncertainty over augmentation (blue hollow triangles).

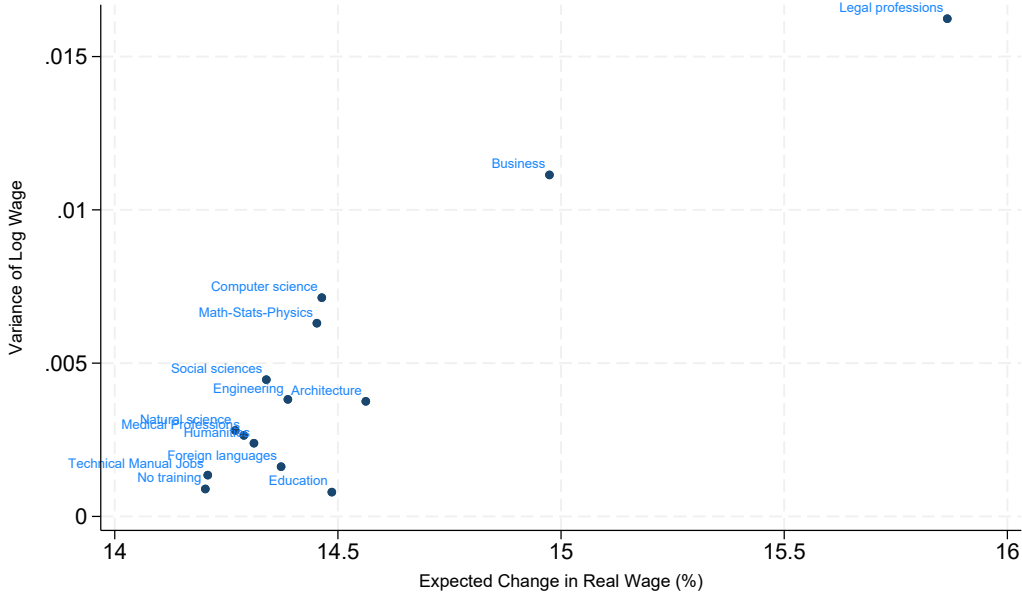


Figure 4: Expected Wage Changes vs. AI Shock Variance across Training Paths.

The model delivers three main results. First, expected automation is the dominant driver of labor reallocation: workers systematically abandon highly exposed cognitive paths—legal professions, business, and computer science—whose employment shares

fall by up to -16% , and crowd into technical manual jobs, education, and the no-training option. Expected augmentation, by contrast, plays a minor role in reallocation because its expected impact and uncertainty are relatively similar across training paths.

Second, the endogenous reallocation of labor generates an across-the-board increase in expected wages of approximately $+14\%$. Because the labor demand curve is downward-sloping, the scarcity of workers in highly exposed paths pushes up their prices and wages, while the congestion in safer paths compresses wages there. These two forces nearly cancel at the aggregate level, yielding a remarkably uniform wage increase that largely offsets the direct negative impact of automation on the most exposed training paths.

Third, automation uncertainty generates a pure general equilibrium risk premium of up to $+2.3\%$. Risk-averse individuals flee volatile training paths independently of their expected returns, creating additional labor scarcity that raises expected wages in those paths above what the expected productivity effects alone would predict. This channel is absent from models that treat AI as a deterministic shock, or that assume symmetric uncertainty across occupations.

Taken together, these results highlight the importance of accounting for the labor supply response to AI. A training path facing severe automation risk may nonetheless see rising expected wages due to the scarcity channel, while a path perceived as safe may experience wage compression from labor market congestion.

References

- Acemoglu, D. and Autor, D. (2011). Skills, tasks and technologies: Implications for employment and earnings. In *Handbook of labor economics*, volume 4.
- Acemoglu, D., Kong, F., and Restrepo, P. (2025). Tasks at work: comparative advantage, technology and labor demand. *Handbook of Labor Economics*, 6:1–114.
- Acemoglu, D. and Restrepo, P. (2018). The race between man and machine: Implications of technology for growth, factor shares, and employment. *American economic review*, 108(6):1488–1542.
- Agrawal, A. K., Gans, J. S., and Goldfarb, A. (2025). Genius on demand: The value of transformative artificial intelligence. Technical report, National Bureau of Economic Research.
- Althoff, L. and Reichardt, H. (2026). Task-specific technical change and comparative advantage.
- Autor, D. and Thompson, N. (2025). Expertise. *Journal of the European Economic Association*, 23(4):1203–1271.
- Autor, D. H. (2019). Work of the past, work of the future. *AEA Papers and Proceedings*, 109:1–32.
- Autor, D. H., Levy, F., and Murnane, R. J. (2003). The skill content of recent technological change: An empirical exploration. *The Quarterly journal of economics*, 118(4):1279–1333.
- Cunha, F. and Heckman, J. (2016). Decomposing trends in inequality in earnings into forecastable and uncertain components. *Journal of labor economics*, 34(S2):S31–S65.
- Cunha, F., Heckman, J., and Navarro, S. (2005). Separating uncertainty from heterogeneity in life cycle earnings. *oxford Economic papers*, 57(2):191–261.
- Eloundou, T., Manning, S., Mishkin, P., and Rock, D. (2024). Gpts are gpts: Labor market impact potential of llms. *Science*, 384(6702):1306–1308.
- Gans, J. S. and Goldfarb, A. (2026). O-ring automation. Technical report, National Bureau of Economic Research.
- Ide, E. and Talamàs, E. (2025). Artificial intelligence in the knowledge economy.
- Levhari, D. and Weiss, Y. (1974). The effect of risk on the investment in human capital. *The American Economic Review*, 64(6):950–963.

- Massenkoff, M. and McCrory, P. (2026). Labor market impacts of ai: A new measure and early evidence.
- McFadden, D. (1972). Conditional logit analysis of qualitative choice behavior.
- Tomlinson, K., Jaffe, S., Wang, W., Counts, S., and Suri, S. (2025). Working with ai: measuring the applicability of generative ai to occupations. *arXiv preprint arXiv:2507.07935*.
- Webb, M. (2019). The impact of artificial intelligence on the labor market. *Available at SSRN 3482150*.